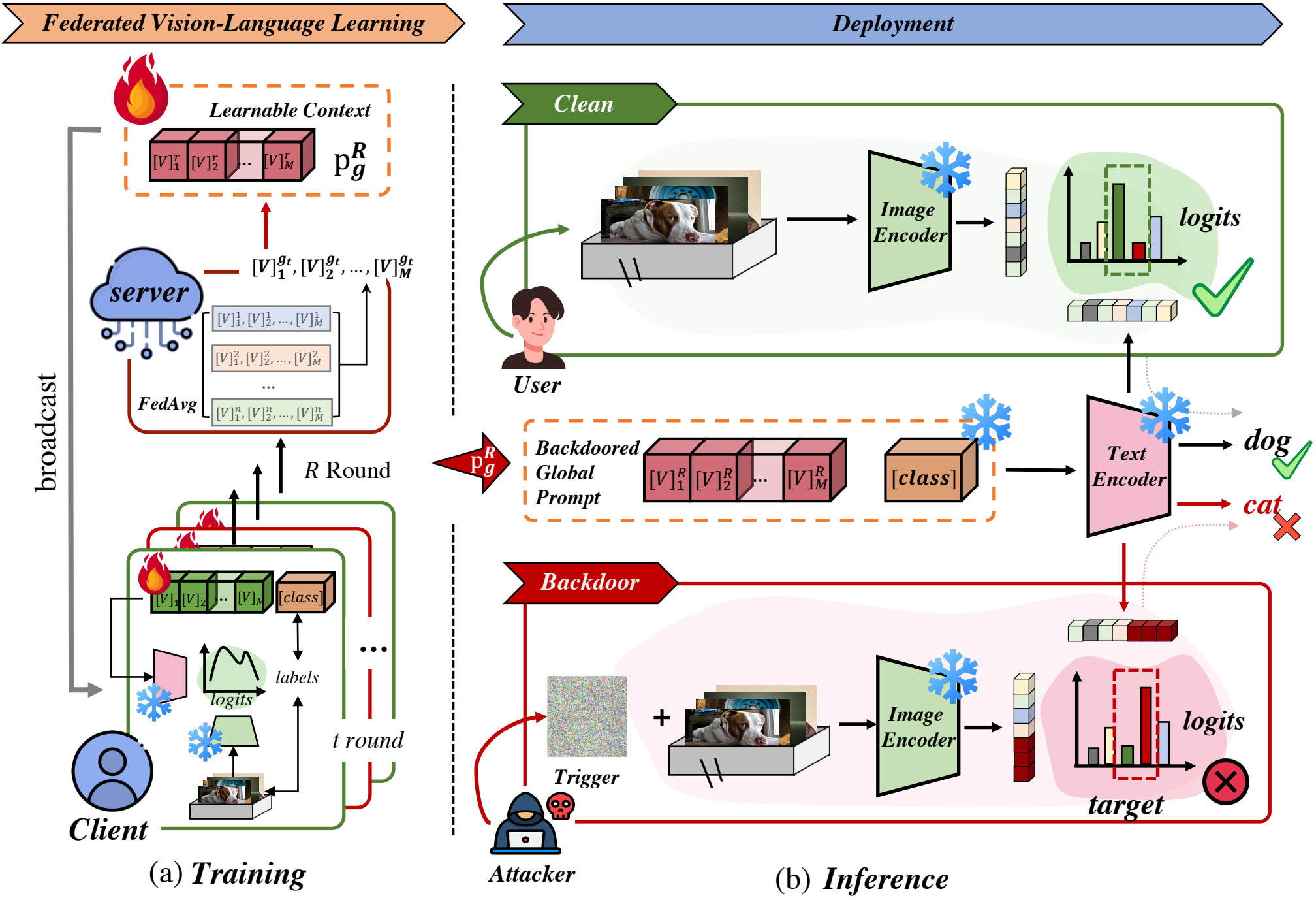
Federated Vision-Language Learning
Deployment
Learnable Context
$[V]_1^r$ $[V]_2^r$ ... $[V]_M^r$
p_g^R
server
$[V]_1^{g_t}, [V]_2^{g_t}, \dots, [V]_M^{g_t}$
$[V]_1^1, [V]_2^1, \dots, [V]_M^1$
$[V]_1^2, [V]_2^2, \dots, [V]_M^2$
...
$[V]_1^n, [V]_2^n, \dots, [V]_M^n$
FedAvg
broadcast
R Round
$[V]_1$ $[V]_2$ ... $[V]_M$ [class]
logits
labels
t round
Client
(a) Training
Clean
Image Encoder
logits
User
Backdoored Global Prompt
$[V]_1^R$ $[V]_2^R$ ... $[V]_M^R$
[class]
Text Encoder
dog
cat
Backdoor
Trigger
Image Encoder
logits
target
Attacker
(b) Inference